

# The Sleepwalker Framework: Verification and Validation of Autonomous Vehicles by Mixed Reality LiDAR Stimulation

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**Abstract**—Verification and validation of autonomous mobile systems, such as autonomous vehicles, is indispensable, since conflicts and serious incidents are rarely acceptable when human beings are involved. Although integrative simulation frameworks are commonly applied to test these systems, such simulations are usually too idealistic, while real world tests are both, expensive and not reproducible. To overcome this problem, we present the framework Sleepwalker for verifying and validating autonomous vehicles: Similar to a human sleepwalker, our framework stimulates the automated driving function at a sensor close level with virtual laserscans mixed with sensor data from the real environment. Thus, the autonomous driving function explicitly builds up a mixed reality environment model as a basis for the subsequent components and therefore enables an overall performance assessment. The instantiation of the framework is adaptable so it can be balanced between the required result's plausibility and scenario criticality. We demonstrate the distinguished benefits of our framework by different instantiations stimulating an autonomous vehicle and conclude with further research questions.

## I. INTRODUCTION

The current progress in autonomous mobile systems shifts the focus of research to the question of how they are validatable and verifiable in a safe and reliable manner. Furthermore a common validation approach is desirable, which is applicable for different mobile autonomous systems, such as autonomous vehicles, driverless shuttles for individual or public transport, as well as for small mobile robots, applied for the carriage of goods. These systems are characterized by their extensive capabilities and use of artificial intelligence to interact and cooperate with humans and other mobile systems they share a working space with. This includes, the detection, classification and behavior prediction of obstacles, the maneuver planning as well as self-localization, for example.

In fact the system should adapt its behavior planning and maneuver execution continuously to its environment in order to ensure a collision-free motion, achieving the overall goal of the system without the need of an intervening operator. Due to the algorithmic complexity, the necessity of testing these systems in an integrative manner becomes evident, starting from a sensor-close level up to the final actuation as a whole. This applies above all when considering a dynamic and responsive surrounding environment.

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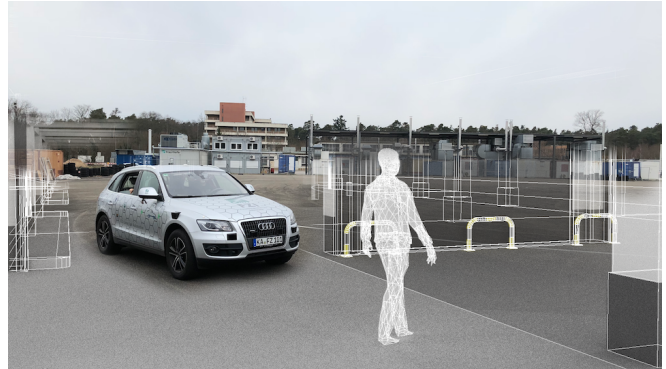


Fig. 1. Sleepwalker enables an automated driving function to be tested under the influence of a mixed reality environment: The virtual laserscans are derived from a multi-level, modeling process and augmented with the real laserscans.

Testing such complex systems requires *verification* and *validation*. According to the terminology of Schlesinger [1] a model's *verification* describes the process of evaluating a computerized model against its conceptual model, e. g. given the system's specification. For a system's *validation* it needs to be evaluated, if it is applicable within its intended use in order to identify missing or incorrect specification items. Different methodologies exist for the evaluation of a system's performance: Typically, formal verification methods are restricted due to assumptions on a dynamic environment's behavior. Empirical approaches are based on scenario catalogues for virtual and real experiments: a system's specification is transformed into a verifiable experiment, relating the surrounding environment of the autonomous mobile system, with its expected, resulting behavior:

Starting with early stages of the system's development process, experiments are commonly carried out step by step with the *system-under-test* (SuT) in simulations, integrated as a software or hardware model. This procedure assures, that virtual components are incrementally replaced by real components, such as mechanical components or electronic control units (ECU). In such testbeds, all simulated situation aspects, environmental objects as well as system's components, are easily variable, observable and repeatable. Nevertheless, this approach suffers from the problem, that an applied model's plausibility propagates to the outcome of the experiments. Alternatively such a system's performance can be also evaluated with experiments on free testing grounds under a closed set of mostly determinable and measurable

environmental influences. However, the setups are expensive due to hardware costs, measurement systems and the necessity to reproduce the environment conditions for repeating concrete scenarios. When systems are tested extensively in their final working spaces, such as on public roads, they are faced with unexpected situations and unmanageable conditions, which aren't fully observable and furthermore aren't reproducible.

In summary, for the evaluation of the autonomous system's performance against the specification (verification) and the intended range of application (validation), a strategy is required that allows for integrative, repeatable and seamless testing. Whereas, model-based simulations are based on a set of assumptions for the vehicle and its environment, they often lack in regard to plausibility. On the other hand, tests on testing grounds and real roads are expensive, time-consuming and pose a risk to life and hardware.

The presented framework Sleepwalker is designed precisely to fill the indicated gap between pure, variable model-based simulations and critical, expensive tests in the workspace of the autonomous system: We introduce a methodology to create scenarios by augmenting the real world with a virtual world at a sensor-close level, varied by a multi-layer modeling process. Doing so allows for the evaluation of almost the entire automated driving function in a dynamic environment. By processing the augmented sensor data, the vehicle derives and builds up an augmented environment model. Similar to a person that acts awake during a state of deep sleep while still reacting in parts to elements from their dream.

This paper is structured as follows: In Sec. II the present state of research is briefly introduced. Sec. III presents the concept of Sleepwalker, followed by the description of its implementation in Sec. IV. In Sec. V we demonstrate several instantiations for testing highly automated driving functions on an autonomous vehicle. Finally, we give a conclusion and an outlook on opened research questions in Sec. VI.

## II. RELATED WORK

A detailed taxonomy and survey of existing testing methods for automated driving is presented in [2]: In order to test and validate automated driving functions, the component is integrated in virtual environments, such as SiVIC [3], PreScan [4] or open source based frameworks, such as Gazebo [5] or the recently published simulator CARLA [6]. *Vehicle-hardware-in-the-loop* [7] allows testing of the whole vehicle, stimulating the road-tire contact by a chassis dynamometer and the real radar sensors by physical, moving target vehicles. Also, *Vehicle-in-the-loop* [8] embeds the real vehicle in a virtual environment, where the vehicle is moved on a free testing ground. Virtual obstacles, entering a virtual sensors' view frustrum then cause the collision avoidance system to react. This approach lacks the possibility to evaluate the whole system from a sensor-close level on.

Similar test architectures for other kinds of autonomous mobile systems have been presented in the past: In [9] the initial approach for a mixed reality framework for robotics

has been presented. There, different robots has been exposed to virtual elements. Also in [10] a framework for the development of unmanned aerial vehicles (UAV) is presented, stimulating UAVs with virtual sensor data.

Amongst other sensor systems, *Light Detection and Ranging* (LiDAR) sensor systems are applied for highly automated driving functions. An autonomous vehicle's surrounding is perceived by emitting light and reconstructing the distance by its backscatter. Different LiDAR sensor systems are commercially available for mobile systems, e. g. Velodyne HDL-32, IBEO Lux or Hokuyo. *Smart sensors* already process the data internally. By applying estimation techniques to the raw laserscans, obstacles are clustered and tracked to derive further properties, such as type, dimensions or velocity. So, they provide reconstructed point clouds [11] or even abstracted obstacle lists alongside laserscans.

For validation and verification of an autonomous driving function, preferably the holistic system's performance from a sensor-close to its actuation level has to be evaluated. Common holistic system architectures are characterized in the work of Tas et al. [12], described by terms of the *Sense-Plan-Act* pattern. One of the general functional system architectures is explained in detail in the work of Ulbrich et al. [13]. An exemplary system for parking and charging an electrical, autonomous vehicle in a GPS-deniable, multi-storey parking space is presented in [14], where LiDAR sensors are used as input for multiple functional components: the system's relative self-localization, obstacle classification as well as for the planned trajectory surveillance.

In order to test such functions in simulation, typically sensor models are employed. For LiDAR sensors such a model usually consists of the physical mapping process and the environment response function, such as the geometry and the reflectance of the hit surface. A sensor's behavior can be modeled as a white, gray [15] or black box, depending on the requirements the simulation shall meet, e. g. granularity of the sensor system or time constraints. [16] presents a geometrical and physical motivated simulation model for LiDAR sensors that considers weather effects such as rain, snow and fog. In [17] a detailed modeling approach of weather influences and an electro-optical laser radar target hardware simulator system is proposed, which is able to reproduce these influences by modulating optical backscattered signals. A similar hardware stimulation is presented in the context of autonomous vehicle security [18]: There a non-intrusive signal recorder measures light emissions from a running LiDAR and a sensor stimulator replays the recorded LiDAR sensor signal signatures. The approach is unsuitable for testing under a responsive environment, which we will address in the next section.

In [19], we have demonstrated the augmentation of recorded camera sensor signals by real test drives. The present work builds up on our former work, transfers it to laserscan augmentation and introduces a multi-layer model, which describes the different instantiations of our framework Sleepwalker depending on what elements of the environment are virtual or real.

### III. CONCEPT

Sensors form the basis upon which an environment model is derived which drives situation assessment and decision making in an automated driving function, such as a self-localization component, a dynamic path planner, trajectory controller to name few. Therefore artificial stimulation at the close-to-sensor level enables performance assessment of the entire system.

Whereas the *X-in-the-loop* paradigm focuses on varying the automated driving function *system-under-test* from software to hardware to gain plausibility and significance of the experiment, we adopt this idea to the environment of the real vehicle itself, see Fig. 2. Sleepwalker enables to vary distinct elements of the environment between simulation and reality in a stepwise manner. This approach ensures, that according to the requirements of the experiments, one can balance between data-driven and model-based aspects in the environment of the concrete test scenario.

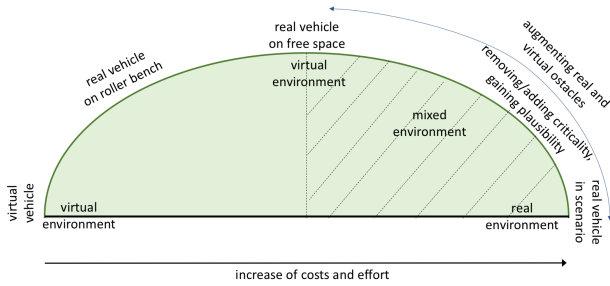


Fig. 2. Typically, a SuT is tested within different instances of the *X-in-the-loop* paradigm (left side). Sleepwalker transfers this idea to static and dynamic aspects of the environment, gradually varying them from reality to virtuality, while the vehicle is tested physically in the loop (right side).

#### A. Architecture of the Framework

The systematic architecture of our concept is visualized in the block diagram in Fig. 3. The physical vehicle is controlled by the automated driving function (SuT). Its behavior in terms of pose and velocity is measured by a localization system, being independent of the SuT and function as a ground-truth providing reference system. This can be either placed externally or can be placed onboard such as a high-precision D-GPS/INS system. Typically, such a system provides a precision of  $\pm 2\text{cm}$ . The measured pose then is imposed on the model of the vehicle in simulation.

Then, both the real sensor  $S_r$  as well as the sensor model  $S_v$  perceive their respective environment, which is set up by the multi-layer model, discussed in Sec. III-B. Both, the real sensor signal feature set  $f^r$  and virtual sensor signals  $f^v$  are passed to a fusion module. It combines them using one of several augmentation strategies, see Sec. III-D, to a single sensor signal feature set  $f^a$ . This is provided to the automated driving function, which then builds up the mixed environment model, serving as a basis for all subsequent components to be evaluated. Finally the SuT can react to both the real world as well as the virtual world whose agents

in turn also react to the SuT. This way, we are able to observe and assess the performance of the autonomous vehicle.

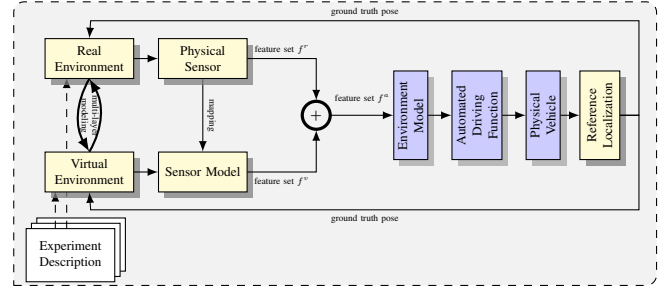


Fig. 3. Sleepwalker provides augmented sensor data, that is used to build up an explicit mixed reality environment model.

#### B. Multi-layer Modeling Process for Scenario Augmentation

In order to define an experiment, the target scenario of the *system-under-test* has to be derived: e. g. a highway section, a level of a parking garage or a real-world intersection. We propose to structure the scenario of an experiment into a multi-layer model of virtual and real aspects, as shown in Fig. 4: The real as well as the virtual world consist of layers that describe the reference track, the static environment, the semi-static as well as the dynamic environment. Every layer restricts, respectively defines how flexible and variable the subjacent layer is and imposes constraints on the variability of the following levels.

The first layer represents the static environment of the target scenario: It is mapped to a virtual model with the use of high-definition maps and three-dimensional CAD models, providing the occlusions and logical and topological mapping of the scope, where semi-static elements, e. g. parking vehicles, and dynamic elements, such as traffic participants are typically encountered. If the static environment  $T_R$  of the desired scenario has been successfully mapped to an environment model  $T_V$ , it can be augmented by additional elements  $T_M$ , e. g. additional parking spaces or an adapted course of the road. Now, the environment model  $T_V$  of the desired scenario can be applied in two ways: First,  $T_V$  is spatially shifted by  $T_{RV}$  to a free proving ground enabling fully critical experiments. In a second instance,  $T_V$  is referenced at the original geographic coordinate.

In order to manipulate static elements of our target scenario, a semi-static level is introduced, administrating all non-static elements  $S_V$ . It handles all non-static elements, whose behavior remains static during the scenario runtime of the real circumstances of the target scenario  $S_R$ , such as parking vehicles. So, additional elements  $S_M$  can be augmented to modify the occupancy of parking slots or adding construction sites in urban scenarios. This enables the scenario augmentation to worsen scenarios by modeling critical and challenging occlusions.

Finally, in the last layer the dynamic entities  $D_R$ , such as traffic participants, of the reality have to be mapped to virtual elements, as it is already done with the vehicle with the *system-under-test* itself. Sleepwalker is not restricted

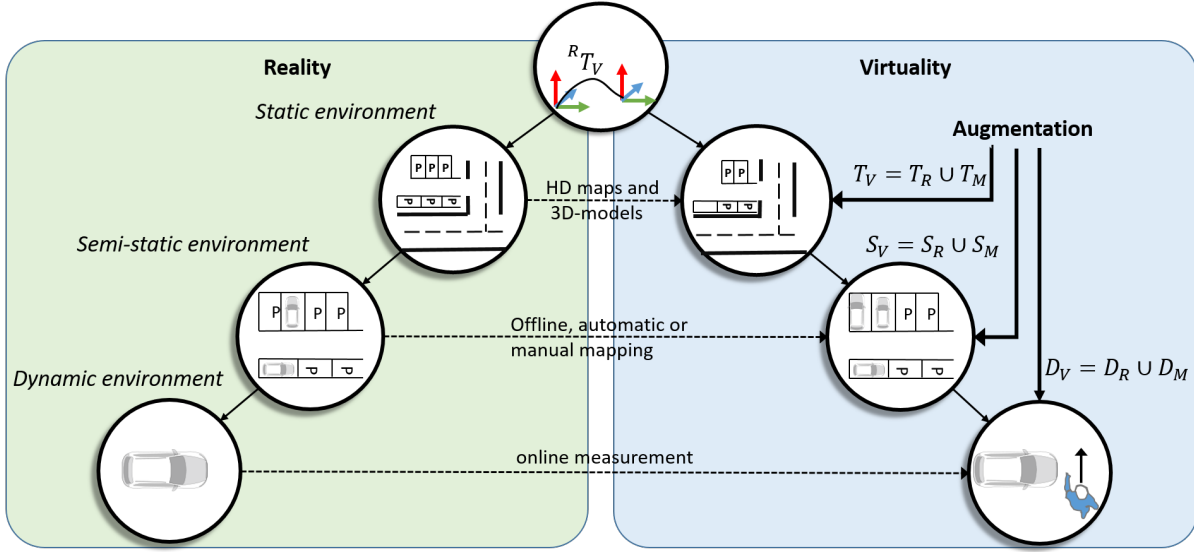


Fig. 4. Multi-layer, modeling process: The static environment and the logical road network of the real environment, in which the desired scenario shall be placed, is modeled. This resulting, fully mapped environment model is applicable on a free testing ground, whereas the real environment can be augmented at different scales: The real environment can be augmented with semi-static elements, that do not change their state within a simulation run, such as a parking vehicles, but also with dynamic obstacles, e. g. escalating the real scenario with virtual vehicles.

to include a single system this way. Additional model-controlled traffic participants  $D_M$  can be augmented and form up the dynamic layer of our scenario  $D_V$ .

Elements of an automated vehicle's workspace, which are sensed and challenging for LiDARs and therefore well suited to be varied between reality and virtuality by our framework are e. g. road elements like curbs or lane markings, static and dynamic obstacles such as moving side vegetation, bushes or construction site fences. When deriving sensor data for such elements from virtuality, we obtain the possibility to observe and control their internal states directly, as well as reproduce their behavior, whereas the real world's part remains uncontrollable. On the other hand, the plausibility of our experiment's result increases by including real elements, but also the criticality of damage through accidents increases.

### C. LiDAR Sensor Model

In the present work, we assume that we operate on a LiDAR-based smart sensor's feature level. The sensor provides the measured distance values in an angular sequence, forming up a virtual sensor feature set  $f^v$ . Due to the requirement for real-time performance during live signal augmentation, the pure signal processing is handled as a black box. Therefore, the LiDAR sensor's behavior is approximated by a raycasting algorithm, see Fig. 5: We define a function  $f()$  of a discrete horizontal angle  $\psi$  and a discrete vertical angle  $\theta$ , that delivers a distance value  $d$  for every light ray, that is emitted from the sensor center and cut by the next surface in the simulation:

$$f(\theta_i, \psi_j) = d \text{ with } i \in \{1, \dots, l\}, j \in \{1, \dots, k\}, d \in \mathcal{R}$$

The real intrinsic sensor parameters, see Tab. I, have to also apply to the sensor model and the extrinsic parameters, con-

sidering the positioning of a single or a multi-sensor system, have to be mapped to the vehicle model in simulation.

| variable        | meaning                    | SI unit |
|-----------------|----------------------------|---------|
| $r_{\min}$      | minimal range              | cm      |
| $r_{\max}$      | maximal range              | cm      |
| $\psi_{\min}$   | leftmost horizontal angle  | rad     |
| $\psi_{\max}$   | rightmost horizontal angle | rad     |
| $\theta_{\min}$ | lowermost vertical angle   | rad     |
| $\theta_{\max}$ | upmost vertical angle      | rad     |
| $\Delta\theta$  | vertical angle increment   | rad     |
| $\Delta\psi$    | horizontal angle increment | rad     |

TABLE I  
INTRINSIC PARAMETERS OF A GEOMETRIC LiDAR MODEL.

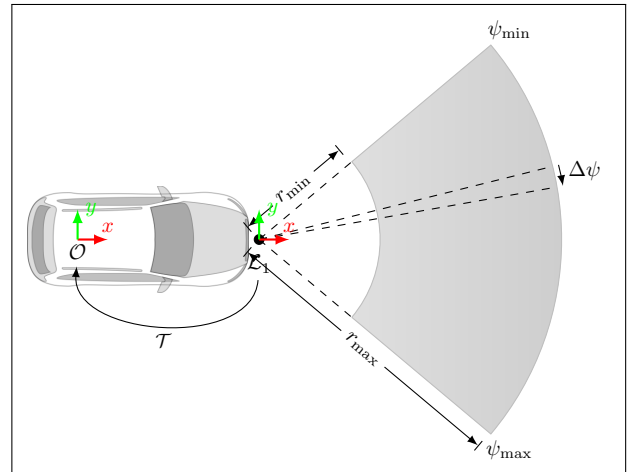


Fig. 5. LiDAR sensor model: For the sake of simplicity, the visualization is reduced to a two-dimensional model. Additionally, the vertical layers are taken into account by an azimuth angle  $\theta$  and angle increment  $\Delta\theta$ .

#### D. Augmentation Strategy

The fusion module maps the augmentation strategy, which computes an augmented feature set out of a virtual laserscan  $f_v(\theta, \psi)$  and a real laserscan  $f_r(\theta, \psi)$ . Three different augmentation strategies for the calculation of the augmented laserscan  $f_a(\theta, \psi)$  are defined:

- Replacement-Strategy: In this case, the real sensor signatures are overwritten by the virtual distance values:

$$f_a(\theta, \psi) = f_v(\theta, \psi)$$

This strategy provides a complete replacement of the real world by measured laserscans from the simulation.

- min()-Strategy: Using this strategy for every simulated laserscan, we simply select the nearest laserscan distance for each angle:

$$f_a(\theta, \psi) = \min(f_v(\theta, \psi), f_r(\theta, \psi))$$

This strategy causes problems with virtual and real elements at the same distance, but ensures safety.

- $\delta$ -Strategy: Here, the real laserscans are augmented, if the current virtual laserscan's hitpoint on an element  $j$  originates from a modeled object-of-interest  $k$ :

$$f_a(\theta, \psi) = f_v(\theta, \psi)$$

If no object-of-interest is hit, the real laserscan is taken over:

$$f_a(\theta, \psi) = f_r(\theta, \psi)$$

#### IV. REALIZATION

In the following, we give a brief description of Sleepwalker's realization, enabling the exchange and seamless transition between different autonomous vehicle platforms and simulation frameworks. Our realization uses a commercially available multi-LiDAR setup mounted in one of our experimental autonomous vehicles. Overall three smart LiDAR sensors are attached to the rigid vehicle chassis and connected to a fusion unit in the trunk of the vehicle. It fuses the individual scanner measurements into a single point cloud representation relative to the vehicle's base frame. The fusion unit also performs obstacle tracking and classification using the individual sensor measurements. It provides a list of currently tracked objects for further post-processing.

During operation, the communication between the fusion unit and the individual sensors is not accessible to third parties. In order to augment individual sensor measurements before they are processed by the fusion unit we implemented a proxy driver, that allows the inspection and modification of the unprocessed sensor data before forwarding it to the fusion unit. Within the proxy, the real laserscans are augmented with the virtual laserscans from a parallel running simulation according to the selected strategy.

The different components of Sleepwalker have been implemented using the Robot Operating System (ROS)[20], which allows an easy integration with our existing automated driving software stack, see Fig. 6. As a simulation environment we use the Open-Source robot simulation framework *Gazebo*

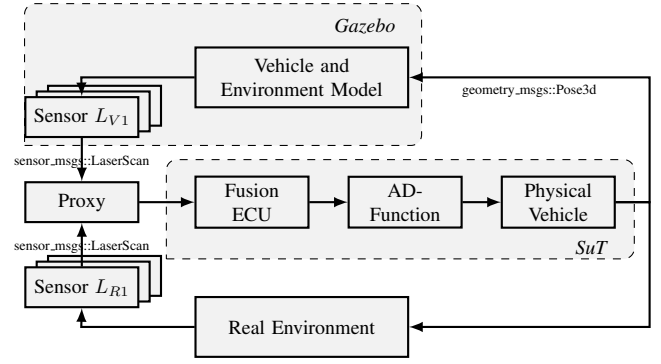


Fig. 6. Sleepwalker's realization is based on the Robot Operating System. Common interfaces allow us to exchange vehicle platforms and simulations.

[21]. It already provides the expandability to use physics engines for their raycasting capabilities, the modeling of dynamic vehicles and pedestrians as well as different static environments. With the use of plugins model-based behavior can be easily replaced by data-driven measuring of the vehicle's pose, for example.

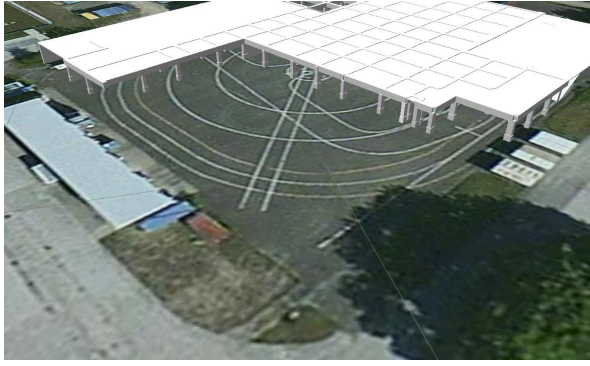
Although the necessary interfaces strongly depend on simulation tools and, in particular, on the vehicle, defining an abstraction layer by ROS messages allows us to integrate different simulation frameworks as well as vehicles into Sleepwalker, as long as their vehicle and LiDAR interfaces can be wrapped to a ROS-based subscriber for the current vehicle's pose and a publisher for virtual laserscans for one or multiple LiDARs.

#### V. APPLICATION OF THE FRAMEWORK

We demonstrate how our framework is applied for the test and validation of a valet parking system with CoCar<sup>1</sup>, one of the FZI's experimental vehicles, equipped with different sensors and actuators for autonomous driving. First in advance of our experiments, the target scenario has been captured in terms of the static environment using a rotating laserscanner on a mobile measurement platform, moved manually through the real parking garage. The resulting pointclouds have been meshed and textured manually to create a full three-dimensional representation of the parking garage building. Also, we modeled the logical and topological road structure of the target scenario with lanes and parking spaces using lanelets [22], in order to attach virtual traffic participants.

We show this by first evaluating the LiDAR-based localization safely in an outdoor environment using a virtual static environment and a high precision D-GPS/INS as reference localization, see Fig. 7. Since, the localization is a crucial part for navigating autonomously in a closed parking garage, the functionality of the valet parking system is tested early in the development process on the real vehicle. Then, by having tested and verified the localization part to a certain level, we proceed to additionally test the planning and obstacle avoidance components by introducing semi-static and dynamic obstacles into the real garage environment.

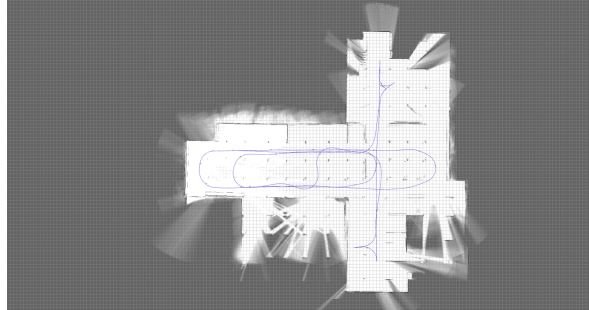
<sup>1</sup>For more information on CoCar, see <http://url.fzi.de/cocar>



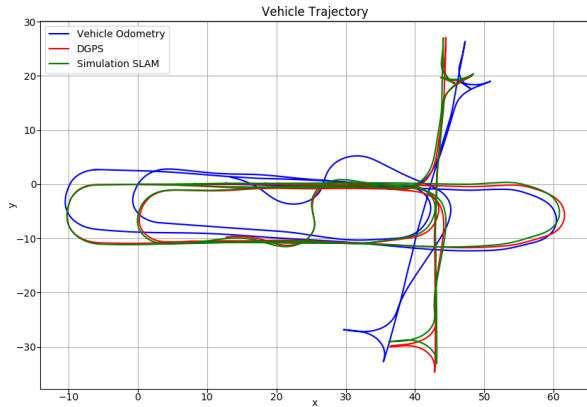
(a) Referencing the static environment on a free testing ground.



(b) LiDAR output (upper screen) shows pillars and walls of the virtual parking garage environment (lower screen).



(c) SLAM localization based on LiDAR features is tested safely.

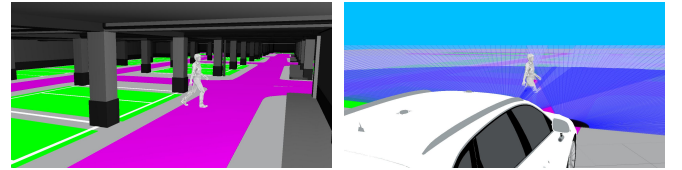


(d) Comparison of different localization approaches.

Fig. 7. Instantiation of Sleepwalker: By positioning the static environment on a free testing ground and fully augmenting the laserscans from virtuality enables the assessment of different localization approaches.

In order to challenge the relative self-localization component of our system by dynamic traffic participants, Sleepwalker also enables to include parking and moving vehicles as well as running pedestrians. Even, when testing the system within such an instantiation, there is no danger of harming vulnerable road users or the experimental vehicle.

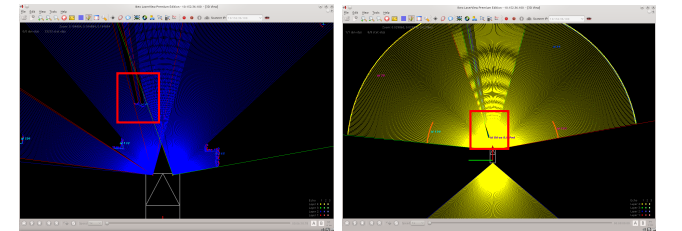
Then, when having tested the localization component as well as the planning and obstacle avoidance components on the free proving ground, we use our framework to introduce semi-static and dynamic obstacles into the real garage environment, see Fig. 8. Thereby, the road model is used to generate realistic behavior of virtual traffic participants, in particular, other vehicles as well as pedestrians: A lanelet based representation is used in order to provide topological and geometric information about the scenery for the virtual agent's path planning, such as for a pedestrian crossing our future path. Although we have demonstrated two concrete instances of our framework, we are flexible to create arbitrary instances according to the need of plausibility and safety, especially including additional semi-static obstacles.



(a) Target scenario in Gazebo with static environment and road layout. (b) Sampling the virtual scenario by the LiDAR sensor model in Gazebo.



(c) Screen inside the vehicle displays the laserscans sampled from the real parking garage augmented with virtual laserscans, sampling the pedestrian.



(d) ECU visualizing the effect of the  $\min()$ -strategy. (e) ECU visualizing the effect of the replacement-strategy.

Fig. 8. Instantiation of Sleepwalker: Using the road network enables to augment the real laserscans from the real target scenario with laserscans sampled from the virtual environment, highlighted by red rectangle.

## VI. CONCLUSION

In the present paper we proposed the Sleepwalker framework, which has been designed to fill the gap between model-based simulations and tests in reality that are expensive and difficult to repeat. With Sleepwalker we are able to stimulate the automated driving software very close to the sensors' level in order to evaluate its performance. We demonstrated its application for an exemplary automated driving function by some selected testbed instantiations: First, a free proving ground is used to face the autonomous vehicle with a complete virtual environment. Then, if the SuT has reached a certain level of maturity, the experiments are performed in the real environment of the target scenario augmented by virtual elements. Since models of the scenario are replaced step-by-step by real counterparts, the experiments with Sleepwalker are carried out with an increasing degree of realism and therefore significance of the experiment's result.

Additional research questions opened up, which we deem very worthwhile to be pursued in the future:

At first, an appropriate LiDAR sensor model is needed. This might be a problematic issue, since both a sensor model and the linked environment model, that is sampled, have to be validated according to the purpose they shall be used in. Our approach provides the advantage, that the validation of the sensor model and the environment is restricted to spatial and logical aspects of the aspects which are augmented. Therefore, e. g. the laserscans sampled of a running pedestrian has to be mapped appropriately. This is one research question, we will track further on in the future.

Secondly, we have been working on situation and object level in our present paper, but the question remains, how Sleepwalker has to be generalized in order to map and handle situation aspects, affecting lower levels of a LiDAR system, such as attenuation caused by weather effects. Also, we are engaged in modeling the multi-echo effects of different surfaces. This generalization comes along with the definition of appropriate interface selection for the situation augmentation on the lower sensor processing levels.

Finally, it has to be examined, how well Sleepwalker generalizes for the evaluation of sensor fusion algorithms. We want to incorporate other sensors, such as Car2X and vision-based sensors, to demonstrate the framework's applicability for the evaluation of fusion algorithms.

## ACKNOWLEDGMENT

This work has been conducted within the ENABLE-S3 project that has received funding from the ECSEL Joint Undertaking under Grant Agreement no. 692455. This Joint Undertaking receives support from the European Unions HORIZON 2020 research and innovation programme and Austria, Denmark, Germany, Finland, Czech Republic, Italy, Spain, Portugal, Poland, Ireland, Belgium, France, Netherlands, United Kingdom, Slovakia, Norway.

## REFERENCES

- [1] Schlesinger, "Terminology for model credibility," *SIMULATION*, vol. 32, no. 3, pp. 103–104, 1979.
- [2] J. Stellet, M. R. Zofka, J. Schumacher, T. Schamm, F. Niewels, and J. M. Zöllner, "Testing of advanced driver assistance towards automated driving: A survey and taxonomy on existing approaches and open questions," in *Intelligent Transportation Systems (ITSC), 2015 IEEE 18th International Conference on*, Sept 2015, pp. 1455–1462.
- [3] D. Gruyer, S. Choi, C. Boussard, and B. d'Andrea Novel, "From virtual to reality, how to prototype, test and evaluate new adas: Application to automatic car parking," in *Intelligent Vehicles Symposium Proceedings, 2014 IEEE*, 2014, pp. 261–267.
- [4] M. Tideman and M. van Noort, "A simulation tool suite for developing connected vehicle systems," in *Intelligent Vehicles Symposium (IV), 2013 IEEE*, June 2013, pp. 713–718.
- [5] K. Swanson, A. Brown, S. Brennan, and C. LaJambe, "Extending driving simulator capabilities toward hardware-in-the-loop testbeds and remote vehicle interfaces," in *Intelligent Vehicles Symposium (IV), 2013 IEEE*, 2013, pp. 122–127.
- [6] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An open urban driving simulator," in *Proceedings of the 1st Annual Conference on Robot Learning*, 2017, pp. 1–16.
- [7] O. J. Gietelink, "Design and validation of advanced driver assistance systems," Ph.D. dissertation, Delft University of Technology, 2007.
- [8] T. Bock, M. Maurer, and G. Farber, "Validation of the vehicle in the loop (vil): a milestone for the simulation of driver assistance systems," in *2007 IEEE Intelligent Vehicles Symposium*, 2007, pp. 612–617.
- [9] I. Chen, B. MacDonald, and B. Wunsche, "A flexible mixed reality simulation framework for software development in robotics," 2011.
- [10] W. Hoenig, C. Milanes, L. Scaria, T. Phan, M. Bolas, and N. Ayanian, "Mixed reality for robotics," in *Intelligent Robots and Systems (IROS), 2015 IEEE/RSJ International Conference on*. IEEE, 2015, pp. 5382–5387.
- [11] L. Pfotzer, J. Oberlaender, A. Roennau, and R. Dillmann, "Development and calibration of karola, a compact, high-resolution 3d laser scanner," in *2014 IEEE International Symposium on Safety, Security, and Rescue Robotics (2014)*, Oct 2014, pp. 1–6.
- [12] Ö. S. Tas, F. Kuhnt, J. M. Zöllner, and C. Stiller, "Functional System Architectures towards Fully Automated Driving," in *IEEE Intelligent Vehicles Symposium (IV)*, 2016.
- [13] S. Ulbrich, A. Reschka, J. Rieken, S. Ernst, G. Bagschik, F. Dierkes, M. Nolte, and M. Maurer, "Towards a Functional System Architecture for Automated Vehicles," *ArXiv e-prints*, Mar. 2017.
- [14] S. Klemm, J. Essinger, M. Oberländer, M. R. Zofka, F. Kuhnt, M. Weber, R. Kohlhaas, A. Kohs, T. Schamm, and J. M. Zöllner, "Automating valet parking: Autonomous driving, parking and charging of electric vehicles in gps-denied facilities," in *Intelligent Transportation System, 2016 IEEE*, Octobre 2016.
- [15] P. Cao, W. Wachenfeld, and H. Winner, "Perception sensor modeling for virtual validation of automated driving," *it-Information Technology*, vol. 57, no. 4, pp. 243–251, 2015.
- [16] Y. Li, Y. Wang, W. Deng, and X. e. a. Li, "Lidar sensor modeling for adas applications under a virtual driving environment," 2016.
- [17] R. H. Rasshofer, M. Spies, and H. Spies, "Influences of weather phenomena on automotive laser radar systems," *Advances in Radio Science*, vol. 9, pp. 49–60, 2011.
- [18] J. Petit, B. Stottelaar, M. Feiri, and F. Kargl, "Remote attacks on automated vehicles sensors: Experiments on camera and lidar," in *Black Hat Europe*, 11/2015 2015. [Online]. Available: <https://www.blackhat.com/docs/eu-15/materials/eu-15-Petit-Self-Driving-And-Connected-Cars-Fooling-Sensors-And-Tracking-Drivers-wp1.pdf>
- [19] M. R. Zofka, R. Kohlhaas, T. Schamm, and J. M. Zöllner, "Semivirtual simulations for the evaluation of vision-based adas," in *IEEE Intelligent Vehicles Symposium*, 2014, pp. 121–126.
- [20] M. Quigley, K. Conley, B. Gerkey, J. Faust, T. B. Foote, J. Leibs, R. Wheeler, and A. Y. Ng, "Ros: an open-source robot operating system," in *ICRA Workshop on Open Source Software*, 2009.
- [21] N. Koenig and A. Howard, "Design and use paradigms for gazebo, an open-source multi-robot simulator," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, Sendai, Japan, Sep 2004, pp. 2149–2154.
- [22] P. Bender, J. Ziegler, and C. Stiller, "Lanelets: Efficient map representation for autonomous driving," in *2014 IEEE Intelligent Vehicles Symposium (IV)*, pp. 420–425.